

Multifunctional Na–CO₂ Electrochemical System: Experimental Validation and Modeling for Sustainable Hydrogen and NaHCO₃ Co-Production

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Abstract

The atmospheric CO₂ concentration has reached an unprecedented level of approximately 427 ppm, contributing significantly to environmental degradation. Existing mitigation techniques are often limited by high costs and low efficiency¹. Electrochemical conversion of CO₂ in aqueous electrolytes offers a promising route for transforming CO₂ into value-added products while mitigating anthropogenic emissions. In the aqueous solution, CO₂ dissolves and produces HCO₃[−], H⁺, and CO₃^{2−}. Continuous purging of CO₂ further decreases the pH, enhancing H⁺ production². In this study, we present a Na – CO₂ system to utilize the produced H⁺ from CO₂ using Na-metal as an anode, an organic electrolyte, NASICON separator, aqueous electrolyte (CO₂ saturated NaOH solution) and Pt/C coated cathode. A Pt/C-coated cathode facilitates the hydrogen evolution reaction (HER), while sodium oxidation occurs at the anode. The Na⁺ ions produced at the anode migrate through the NASICON separator and react with bicarbonate ions (HCO₃[−]) in the catholyte to form sodium bicarbonate (NaHCO₃). This process enables the simultaneous production of hydrogen gas, sodium bicarbonate, and electrical energy.

A multiphase transient mathematical model using dilute solution theory, binary electrolyte theory, and laminar bubble flow in aqueous electrolyte is developed and validated against experimental results. The model provides insights into the system's behavior by resolving concentration fields, reaction zones, local currents, and potentials. Additionally, a parametric study is performed to assess the impact of key design parameters, including cathode–separator distance, anode size, electrode porosity, and chemical reaction rate constant. This combined experimental and modeling approach offers comprehensive insights into the operation and optimization of Na – CO₂ electrochemical systems for sustainable energy and CO₂ utilization.

Keywords: Na-CO₂ battery, CO₂ utilization, Energy Conversion, Bubbly flow, Hydrogen Production

References

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